Data Types and Information Purpose

Understanding data and its importance

Transactional Data

Understanding daily business actions and records

Daily Actions

Transactional data is the data of daily business activities. It records what happens every day.

Examples:

- Trading company: buying and selling records
- Manufacturing company: production activity records
- Ride-hailing / cab company: trip records (pickup, drop-off, time, fare)
- HR basics: hiring and firing records

Key point:

Transactional data is usually very large because it is created all the time.

It is stored in operational systems such as an ERP system.

Master Data

Core information for business transactions

People Data

Master data is the main “core” information that transactional data uses.

Cab company example:

- Transactional data: one trip record
- Master data inside the trip: driver details, passenger details, locations, and base fare

Common master data includes:

- People: customers, suppliers, employees
- Places: addresses, cities, countries
- Things: products, assets, items

Key point:

Master data is often managed in a specific system:

- Employee master data → HR system
- Product master data → inventory system

Reference Data

Understanding standard lists for consistency

Standard Lists

Reference data is a small standard list used to keep data consistent.
It is part of master data, but it changes very rarely.
Examples:

- List of countries (standard country codes)
- Currency codes
- Status lists (Active / Inactive)

Cab company example:

- Driver list can change often (master data)
- Country list almost never changes (reference data)

Reporting Data

Understanding the role of aggregated data

Aggregated Data

RepoReporting data is summarised (aggregated) data used for reports and analysis.
It can include transactional + master + reference data together.
Example:

- "Trips on 13 July in Greater London"
  - Trips = transactional + master
  - Greater London region = reference

Key point:
Reporting data supports planning and decision-making.
Reporting data combines multiple data sources for clarity.

Metadata

Understanding data about data in detail

Definition

Metadata means “data about data.”
It gives extra details that help explain the data.
Example (photo):

- You can see the cat is grey (main data)
- But metadata tells you:
  - when the photo was taken
  - where it was taken
  - camera details

Key point:
Metadata answers questions you cannot answer by looking at the main data only.

Purpose of Information

Understanding why information is crucial for organisations

Analysis

Information helps track **sales trends** and performance changes.

Daily Operations

Information supports routine tasks like **scheduling staff** effectively.

Decision Making

Data-driven insights help in making **strategic business choices**.

Commercial Use

Collaborating with partners enables **joint promotional strategies**.

Purpose of Analysis

Understanding data's role in organisational success

Information helps businesses analyse performance.

Examples of questions:

- Is profit increasing or decreasing?
- Which marketing plan worked better?
- Should we use this marketing plan again?

Example:

A company compares sales between stores to see which branch is doing well or badly.

Support Daily Operations

Information helps employees do their work.

Staff Schedules

Organised schedules ensure teams are present when needed.

Task Reminders

Alerts help employees remember important deadlines and tasks.

Stock Checks

Regular checks prevent shortages and ensure product availability.

Timing in Food

Accurate timing improves service speed and customer satisfaction.

Decision Making

Using data to guide important choices

Information helps managers make important decisions using:

- sales figures
- costs
- profit margins

Example:

If a store is performing badly, managers may decide to:

- reduce staff hours
- change prices
- improve profit margins
- or even close the store

Commercial Use (Working with other businesses)

Leveraging data to enhance partnerships and events

Businesses use information to plan events and cooperate with partners.

Example:
A hotel runs a wedding event and works with local dress and cake shops.
They can offer special promotions to attract customers.

Information Sources

Understanding data sources in organisations

External Sources

Includes laws, tax rules, and market reports.

Internal Sources

Comprises HR, finance, marketing, and purchasing data.

External Sources (Outside the organisation)

- Government rules and laws (tax, VAT, pensions, national insurance)
- Market reports
- Paid information providers (information traders)

Important note:
Businesses must follow data laws when storing and using customer information.

Internal Sources (Inside the organisation)

Different departments create different information:

- HR / Personnel: hiring, interviews, employee records
- Finance: invoices, payroll, VAT, accounts
- Marketing: target audience, market research, advertising designs
- Purchasing: suppliers, prices, contracts
- Customer Service / Sales: FAQs, client lists, contact details.
- Manufacturing: designs, maintenance records, production records
- R&D: research, customer feedback, product improvements